

Mikhail Razakov

C++ / LLVM Developer

JUNIOR / INTERNSHIP

C++23 · LLVM · program analysis · x86-64

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PROFILE

I build systems and compiler components: tested C++23 graph algorithms, conservative memory-dependence analysis, and an experimental x86-64 backend.

KEY EVIDENCE

1st of 49

1st among 49 submitted solutions; the only 5,0/5,0 score and 104/104 official tests in the LLVM compiler track selection.

yes / no / maybe

Conservative memory-dependence analysis with exact affine analysis and cost-guarded fallbacks.

Correctness verification

Exact-oracle, differential, and metamorphic testing, replay/reduction, and conservative fail-closed decisions.

EXPERIENCE

MCST — Compiler Engineering Intern

July — August 2026

LLVM · C++23 · remote

- Developing an LLVM optimization pass.
- Implemented cycle-detecting RPO, Dijkstra, Dinic maximum flow, and Tarjan SCC over a shared C++23 graph model; CMake build with input/output checks and a Python test runner.

[GitHub](#)

SELECTED PROJECTS

PS-form Memory Dependence Analyzer

C17 · program analysis

- C17 analyzer for parametric memory-access dependence with conservative yes/no/maybe outcomes.
- Symbolic normalization, residue/GCD abstractions, exact affine analysis, and cost-guarded fallbacks; a harness with an exact oracle, metamorphic WA hunting, and model-guided TL hunting.

[GitHub](#)

x86-64 Codegen & Register Allocation Playground

Rust · x86-64 · register allocation

- Experimental playground for x86-64 code generation, linear-scan register allocation, and reproducible correctness/performance checks.
- SSA-like IR, liveness, assignment verification, and differential parity with an IR interpreter; explicitly not positioned as a production-ready backend.

[GitHub](#)

SKILLS

Toolchain	C++23, C17, CMake, Linux, clang-tidy, ASan/UBSan
Analysis	CFG, dominance, SSA, SCC, dataflow, memory dependence
Codegen	x86-64, liveness, linear scan, IR interpretation
Verification	exact oracles, differential/metamorphic tests, stress testing, reduction

EDUCATION AND RECOGNITION

HSE University — Software Engineering program, since 2026.

HSE Vysshaya Proba prize-winner in competitive and industrial programming; two-time first-degree diploma recipient at the MEPhI Junior competition.